Rahul Sajnani | rahul_sajnani@brown.edu | Homepage | Scholar | GitHub | LinkedIn

Education

- **Ph.D. in Computer Science**, Brown University • Advisor: Prof. Srinath Sridhar • CGPA 4.0/4.0 — Sep 2022 – Dec 2026
- **B.Tech. & M.Sc. in ECE**, IIIT Hyderabad • Gold Medal, Dean's List • CGPA 9.1/10 — Aug 2017 – May 2022

Research Experience

Amazon Robotics — 2024 & 2026

Applied Scientist II Intern, North Reading, MA — *Managers: Jeroen Vanbaar, Jose Barreiros, Kapil Katyal, Manikantan Nambi*

- **Designed and trained UniWALM**, a unified world-action-language model (WAM/VLA) for memory-dependent robot manipulation — *reasoning at the speed of action*. Introduced a Mixture-of-Transformers backbone that co-generates future video, action, and language in a **single parallel diffusion pass** via a **discrete-diffusion language model (dLLM)**; the policy re-reads both as a **visual + textual memory + reasoning loop** on every control step — so the robot remembers the past and reasons about future subtasks. **Designed** the bimanual and memory-dependent data-collection protocols and **deployed UniWALM with memory** on Sharpa Wave, Paxini hand, Flexion arms, Franka Panda, and AgileX dual-arm. **Raised real-world success** 11.1% → 61.1% vs same-backbone baseline; reached **34.5**% on RMBench, beating Mem-0's privileged-keyframe setup (28.5%).
- **Built GeoDiffuser**, a training-free 2D/3D image editor on a frozen T2I diffusion model (**WACV 2025 Best Student Paper**, invited talk) that realizes user-specified 3D rigid transformations by rewriting the model's attention maps — no fine-tuning. Used it to **generate realistic tote-object data** for robot manipulation training.

Adobe Research — May 2025 – Aug 2025

Research Scientist Intern, San Jose, CA — *Advisors: Matheus Gadelha, Radomír Měch*

- **Built AppearancePointers**, a FLUX-based image generator that steers each output region independently by text, reference image, or material texture in a single denoising pass. **State-of-the-art** on three multi-region fidelity metrics — beats DreamRenderer and MSDiffusion at larger scale; curated the AP-37K multimodal dataset.

Brown Visual Computing, Brown University — Sep 2022 – Present

Graduate Researcher, Providence, RI — *Advisors: Srinath Sridhar & Leonidas Guibas*

- **Built** self-supervised SO(3)-equivariant 3D canonicalization for partial point clouds (ConDor, CVPR 2022) and NeRFs (Cafi-Net, CVPR 2023 Highlight), and a diffusion-based scene-rearrangement model (LEGO-Net, CVPR 2023). Currently building a feed-forward **3D Gaussian Splatting** model that reconstructs a scene from a single monocular video, by distilling pretrained video-diffusion backbones (Wan2.1, LTX-Video, CogVideoX).

Robotics Research Center (RRC), IIIT Hyderabad — Aug 2019 – May 2022

Undergraduate Researcher, Hyderabad, India — *Advisor: Prof. K. Madhava Krishna*

- **Built DRACO** (ICRA 2021), a weakly-supervised method for dense object-pose canonicalization that works without 3D ground-truth, and a multi-object monocular SLAM for dynamic driving scenes (IEEE IV 2020); **designed** mapping and obstacle-avoidance modules for an autonomous self-driving car [Video].

Publications

UniWALM: Unified World-Action-Language Model for Embodied Robot Control. *Under Review @ ICRA 27* [Project Page] [arXiv] 2026
Rahul Sajnani, Diego Romeres, Mingyuan Zhang, Kai Gao, Carl Qi, Baichuan Huang, Srinath Sridhar, Dylan Randle, Lillian J. Ratliff, Jose Barreiros, Manikantan Nambi, Jeroen Vanbaar.

AppearancePointers: Fine-Grained Multi-Region Control with Appearance Pointers. *Under Review @ ICLR 27* [Project Page] [arXiv] 2026
Rahul Sajnani, Matheus Gadelha, Yulia Gryaditskaya, Srinath Sridhar, Radomír Měch.

AnchorQ: Sample-Efficient Reinforcement Learning for Policy Improvement via Flow-Reversed Action Anchors. *Under Review @ ICLR 27* 2026
[Project Page] [arXiv] Carl Qi, Weijun Dong, Xiaojie Wang, Peiyi Chen, **Rahul Sajnani**, Mingyuan Zhang, Amy Zhang, Jose Barreiros, Manikantan Nambi, Lillian J. Ratliff, Dylan Randle.

GeoDiffuser: Geometry-Based Image Editing with Diffusion Models. *IEEE WACV 2025 (Best Student Paper)* [Project Page] [Code] 2025
Rahul Sajnani, Jeroen Vanbaar, Jie Min, Kapil Katyal, Srinath Sridhar.

GenHSI: Training-Free Controllable Generation of Human-Scene Interaction Videos. *IEEE WACV 2026* [Project Page] [arXiv] 2026
Zekun Li, Rui Zhou, **Rahul Sajnani**, Xiaoyan Cong, Daniel Ritchie, Srinath Sridhar.

ConDor: Self-Supervised Canonicalization of 3D Pose for Partial Shapes. *IEEE CVPR 2022* [Project Page] [Code] 2022
Rahul Sajnani, Adrien Poulenard, Jivitesh Jain, Radhika Dua, Leonidas J. Guibas, Srinath Sridhar.

Canonical Fields: Self-Supervised Learning of Pose-Canonicalized Neural Fields. *IEEE CVPR 2023 (Highlight)* [Project Page] [Code] 2023
Rohith Agaram, Shaurya Dewan, **Rahul Sajnani**, Adrien Poulenard, Madhava Krishna, Srinath Sridhar.

LEGO-Net: Learning Regular Rearrangements of Objects in Rooms. *IEEE CVPR 2023* [Project Page] 2023
Qiuhong Anna Wei, Sijie Ding, Jeong Joon Park, **Rahul Sajnani**, Adrien Poulenard, Srinath Sridhar, Leonidas Guibas.

DRACO: Weakly Supervised Dense Reconstruction And Canonicalization of Objects. *IEEE ICRA 2021* [arXiv] 2021
Rahul Sajnani, AadilMehdi Sanchawala, Krishna Murthy Jatavallabhula, Srinath Sridhar, K. Madhava Krishna.

Skills, Awards & Service

Skills: PyTorch · FSDP/DDP · Hydra · W&B · Transformers/Attention · Pretraining & Fine-tuning · Flow Matching · LoRA · Behavior Cloning · Imitation Learning · Diffusion Policy · Discrete-Diffusion LM (dLLM) · VLMs/LLMs · Self-Driving · PyTorch3D · ROS · FastAPI · React · Claude Code · Python · C/C++. **Awards:** **WACV 2025 Best Student Paper** (GeoDiffuser, invited talk at WACV); **Gold Medal**, IIIT Hyderabad (2022); **CVPR 2025 Outstanding Reviewer**. **Service:** Co-Instructor Topics in 3D CV & ML (CSCI 2952K), Brown (2025); Head Organizer Robotics Summer School, IIIT-H (2021); Reviewer: CVPR, ICCV, ECCV, ICLR, SIGGRAPH.